Problem Chosen	2023 MCM/ICM Summary Sheet	Team Control Number
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GGDP:replaces GDP and makes the world a better place

Summary

"Heal the world, make it a better place. For you and for me, and the entire human race". Nowadays, Human race is running out of the wealth for future generations. Fortunately, a new economic measurement, Green GDP (GGDP), might replace GDP and "heal" the world. To better understand GGDP and corresponding shift policy (from GDP to GGDP), we analyzed this policy's impact on climate mitigation, its upsides and downsides, and particularly, analyzed its implementation in China.

To achieve above objectives, three models are established.

For **Model 1**: We choose the **adjusted-GDP** method to compute GGDP. We recognize the difference between GGDP and GDP according to Integrated Environmental and Economic Accounting (IEEA), and divide the difference into three parts: **resource consumption, environment pollution cost** and **environment governance cost**. Additionally, for each part, we find as much as specific indicators to represent them. In the end, we formulate a simplified GGDP-computing model which uses accessible data but also accurately represents the GGDP.For **Model 2**: An index (called c_{index}) measuring the climate mitigation was built. It is composed of **carbon emission, greenhouse emission** and proportion of renewable resource in the whole resource consumption. The weight of each indicator (**0.34, 0.51, 0.15**) was computed by **Entropy Weighting Method**. We then collected data and computed the GGDP, used c_{index} as dependent variable and GGDP as independent variable, and built the **GGDP-Impact-on-Climate model** by linear regression method. The result shows that an increase in GGDP (1 trillion increase) will bring a **-1.63** impact on the index. We also did the same on GDP, and it shows an increase in GDP will bring a -1.70 impact on the index. Combined these two results, setting the GGDP as economic measurement does have a positive impact on climate mitigation.For **Model 3**, We collected and used 2006-2019 panel data on China's industrial provinces to analyze the impact of environmental policies on **green total factor productivity (GTFP)**, using a **dynamic policy measurement model** based on the Systematic Generalized Method of Moments (SGMM). We found that a mix of three types of environmental policies, which is a specific change of China when using GGDP, can achieve high-quality and sustainable economic development in the short and long term in China.

To test Model 2, its R-squared is computed and the value is 0.9614, which shows it is a good fitting result. Another model with GDP as independent variable also shows a good fitting (its **R-squared** is 0.9567). To test Model 3, a **robustness test** and a **stress test** were performed. Both the tests show that the model is reliable(The numerical result is displayed in the main body of article).To sum up, in this article we prove the policy of shifting economic measurement from GDP to GGDP is beneficial both in the short and long run. In addition, the models we established are proved to be reliable and potential, which could be used in future policy analysis.

Keywords: GGDP, climate mitigation index, linear regression, SGMM

Dear Chinese governors,

I am writing to recommend a significant shift in the way we measure economic development, namely from GDP to green GDP (GGDP).

"Lucid waters and lush mountains are invaluable assets." China's economy has achieved amazing growth over the past few decades, but this growth has come at the cost of high pollution, high energy consumption, and high waste. It’s time to change the standard for measuring economic development. GGDP can be a good choice which reflects the balance between economic development and environmental protection.

GGDP can help promote the implementation of China's sustainable development strategy. China has incorporated sustainable development into its overall national development plan, and GGDP can better reflect the requirements of sustainable development. Secondly, GGDP will encourage China's economic restructuring and upgrading. Currently, China's economic growth relies mainly on energy-intensive and resource-intensive industries. However, shifting to GGDP will guide enterprises to transform and upgrade towards environmentally friendly industries. Moreover, GGDP can help improve China's influence in global environmental governance. Nowadays, the green economy has become an important strategy for promoting sustainable development globally. If China trusts GGDP, it will help improve China's status and influence in global environmental governance. Finally, shifting to GGDP can provide a great impetus for technological innovation and progress.

Regarding the conversion cost of GGDP, it is true that the process may bring some economic burden to the country. However, in the current economic situation in China, the conversion cost of GGDP can be offset through measures such as reducing environmental governance costs and improving resource efficiency. According to statistics, the cost of environmental governance in China already accounts for more than 2% of GDP annually, and the impact of environmental pollution on people's health and production is incalculable. Hence, the conversion to GGDP can achieve benefits by reducing environmental governance costs.

Therefore, I strongly recommend that the Chinese government switch from GDP to GGDP. This is a major decision that is in line with the historical trend and has global foresight, as well as an important measure for China to build an ecological civilization and promote sustainable development.

Sincerely,

Your loyal followers

Contents

1 Introduction **4**
- 1.1 Problem Background . . . 4
- 1.2 Restatement of the Problem . . . 4
- 1.3 Literature Review . . . 5
- 1.4 Our Work . . . 6

2 Assumptions and Justification **6**

3 Notations **7**

4 Data Collection and Analysis **7**

5 A simplified model of GGDP **7**
- 5.1 Comparison of different calculation methods . . . 7
- 5.2 Model establishment . . . 8
- 5.3 Comparative analysis of GGDP and GDP . . . 10

6 A GGDP-Impact-on-Climate Model **11**
- 6.1 A synthetic indicator of climate mitigation . . . 11
- 6.2 Model establishment . . . 12
- 6.3 Model Solution . . . 12
- 6.4 Comparison with GDP . . . 13

7 Potential Upside and Downside During Conversion **14**
- 7.1 Potential income analysis . . . 14
- 7.2 Potential cost analysis . . . 15
- 7.3 Comparison and Summary . . . 15

8 Concrete analysis of switch from GDP to GGDP **16**
- 8.1 Specific situation in China . . . 16
- 8.2 Specific Changes Caused by Switch . . . 16
- 8.3 Effect analysis . . . 17
- 8.4 Conclusion . . . 19

9 Model Test——Robustness test and Stress test **20**

10 Model Evaluation **21**
- 10.1 Strength . . . 21
- 10.2 Weakness . . . 21

Reference **22**

1 Introduction

1.1 Problem Background

Gross Domestic Product (GDP) is a widely used measure of a nation's economic health, used for determining purchasing power and access to loans. However, GDP fails to consider the conservation of natural resources and environmental sustainability, resulting in policies and projects that prioritize immediate production over long-term environmental impacts. This is exemplified by actions such as clear-cutting forests and overfishing, which boost GDP without regard for the loss of biodiversity and negative environmental consequences. The proposal of a "Green" GDP (GGDP), which includes environmental and sustainability perspectives and factors, could promote policies and projects that are better for the planet's environmental health. Although convincing nations to shift from conventional GDP to GGDP as the primary measure of economic health would be challenging, if it initiates a global movement towards climate crisis mitigation, it may be worth the effort.

1.2 Restatement of the Problem

GDP is a limited measure of a country's economic health as it fails to account for environmental impacts, such as degradation, resource depletion, and other factors that contribute to long-term well-being. However, as a widely recognized indicator of economic development, GDP has been accepted and used worldwide for many years. The important issue is how to effectively promote green GDP (GGDP) and have it accepted by countries as a measure of their development. This is a critical proposition for promoting global sustainable development. Therefore, based on the requirements of the question, we need to address the following problems:

- **Problem 1** Currently, there are many proposed methods for calculating GGDP, and the task of this question is to select a method that can effectively measure progress towards climate goals and can be implemented in practice. By choosing a method that can effectively mitigate climate change, countries can better align their economic policies with climate goals and work towards a more sustainable future.
- **Problem 2** We need to design a simple model to evaluate the expected global impact of the GGDP indicator. In addition, the question requires us to decide how to measure the impact on the global climate.
- **Problem 3** Analyze the potential benefits and costs and efforts required for replacing GDP with GGDP globally. Based on the analysis and the results of the model, weigh the potential benefits and costs and support the plan to replace GDP with GGDP as the measure of economic development.
- **Problem 4** Choose a country and conduct a more in-depth analysis of the possible effects of adopting GGDP. Explore the specific changes in how the country uses or conserves its natural resources under current GDP and with the adoption of GGDP. Also, consider whether these changes are beneficial for the country in terms of its current economic status and future development capacity.
- **Problem 5** After conducting a country-specific analysis, the task requires us to prepare a one-page report in simple language, free of technical jargon, to outline whether it is recommended

for the leaders of that country to adopt GGDP as the primary measure of national economic health or to continue using GDP as the existing measure.

1.3 Literature Review

1.3.1 Research on GGDP Calculation Methods

Currently, scholars in various countries have proposed multiple methods for calculating GGDP, and multidimensional research has explored different calculation methods.

One method for calculating GGDP is based on the Genuine Progress Indicator (GPI), which considers factors such as income distribution, non-market services, and environmental costs and benefits. Lawn and Clarke (2010)[**1**] proposed this method and calculated Australia's GGDP, finding that the GGDP growth rate was lower than the traditional GDP growth rate. Another study by Göransson et al. (2015)[**2**] used the GPI method to calculate Sweden's GGDP and found that the GGDP growth rate was higher than the GDP growth rate, indicating that Sweden's economic growth was accompanied by an improvement in environmental quality.

Cordero and Davidsdottir (2019)[**3**] based their research on the ecological footprint (EF), measuring the land and water resources needed to maintain human activities and absorb waste, and calculated Iceland's GGDP. They found that the GGDP growth rate was lower than the GDP growth rate, indicating that Iceland's economic growth came at the expense of the environment. Liu and Sun (2018)[**4**] used the carbon footprint (CF) to measure the amount of carbon emissions generated by economic activities and calculated China's GGDP. They found that the GGDP growth rate was higher than the GDP growth rate, indicating that China's economic growth was accompanied by an increase in carbon emissions.

Overall, these studies indicate that different methods for calculating GGDP may yield different results. Further research is needed to determine which method or combination of methods is most suitable for measuring economic growth while considering environmental sustainability.

1.3.2 The relationship between GGDP and climate mitigation

In a study by Zhang et al. (2019)[**5**], a panel data regression model was used to investigate the relationship between GGDP and carbon emissions. The results showed that increasing GGDP is associated with reducing carbon emissions, indicating that adopting GGDP as the primary measure of economic performance can promote climate mitigation. Another study by Wang and Liu (2018)[**6**] analyzed the relationship between GGDP and air pollution in China. The study found that increasing GGDP is associated with reducing air pollution, indicating that GGDP can also promote environmental sustainability. A study by Ang and Xu (2011)[**7**] showed that incorporating environmental factors into economic policies can achieve more sustainable and low-carbon economic growth. Some studies suggest that using GGDP as the primary economic indicator would lead to more sustainable economic growth and promote efforts to mitigate climate change. A study by Miao et al. (2020)[**8**] showed that transitioning to a GGDP-based economic system could lead to a more sustainable and lower-carbon economy.

1.4 Our Work

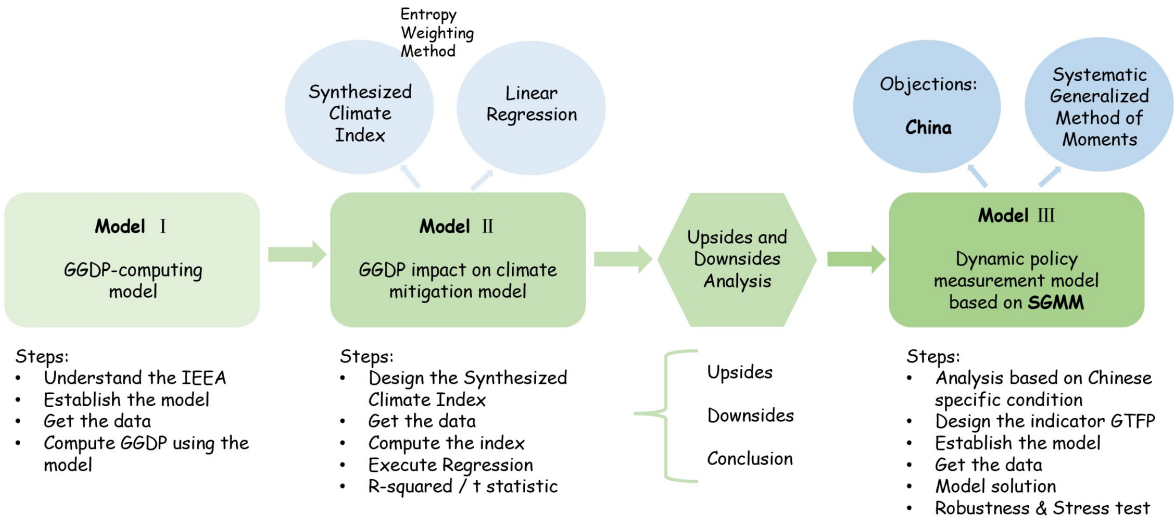


Figure 1: Work Flow

2 Assumptions and Justification

- We assume that the depletion of renewable resources is the same as that of non-renewable resources. When calculating resource depletion, the depletion of renewable resources is not specifically considered.

  Justification: Although renewable resources can gradually regenerate and replenish the depletion, this process is very slow and requires a long cycle. Moreover, currently, the self-replenishment speed of most renewable resources is far lower than the consumption speed, so it can be approximated that no replenishment is taking place.

- We assume that the cost of environmental pollution losses does not include the losses caused by economic activities that cannot be carried out due to environmental pollution.

  Justification: Although in theory, these economic activities that cannot be carried out due to environmental pollution can be considered as losses caused by environmental pollution, it is difficult to obtain data on these activities and the judgment criteria are subjective, so they are not considered.

- We assume that the cost of diseases caused by environmental pollution to the human body can be regarded as indirect losses caused by environmental pollution.

  Justification: The harm caused by environmental pollution to the human body is becoming more and more well-known, and studies have found that the cost of such losses is rising at a rapid pace. Therefore, it should be taken into consideration.

3 Notations

Symbols	Description	Unit
x_{GDP}	represents yearly GDP value in dollars	$
x_{GGDP}	represents yearly GGDP value in dollars	$
M_1	Represents annual carbon emissions	kt
M_2	Represents annual greenhouse gas emissions	kt
M_3	Represents the proportion of renewable energy consumption to total energy consumption	%
c_{index}	a synthesized index represents the climate mitigation computed from M_1, M_2, M_3	-
Y_{cm}	the same as c_{index}	-
$SGTFP_{it}$	Represents the short-term level of green total factor productivity for province i in year t	-
$MGTFP_{it}$	Represents the medium-term level of green total factor productivity for province i in year t	-
LGTFP$_{it}$	Represents the long-term level of green total factor productivity for province i in year t	-
IER	Investment-oriented regulatory intensity	-
FER	Cost-oriented environmental regulatory intensity	-
Sub	Investment-oriented regulatory intensity	-
\$Y$_{it}$\$	Represents the calculated value of green total factor productivity for province i in year t	-

4 Data Collection and Analysis

This article uses relevant global data from 2010 to 2020, sourced from the World Bank, Eurostat, International Energy Agency, National Environmental Protection Department, United Nations Environment Programme, Global Disaster Database, Global Mineral Resource Information Platform, World Water Development Report, BP Statistical Review of World Energy, and Environmental Sustainability Index Report.

5 A simplified model of GGDP

5.1 Comparison of different calculation methods

After reviewing relevant literature and materials, we have found that there are several methods for calculating GGDP. These include:

1) Adjusted GDP: This method involves adjusting traditional GDP to include costs related to environmental degradation, such as pollution and natural resource depletion. Environmental damage costs are subtracted from GDP to obtain an adjusted value.

2) Genuine Progress Indicator (GPI): This is an alternative measure to GDP that considers social and environmental factors in addition to economic factors. GPI takes into account the positive and negative impacts of economic activity on people's well-being and the environment, including the value of unpaid labor, income inequality, and environmental degradation.

3) Ecological Footprint (EF): This method evaluates the environmental impact of economic activities by measuring the land, water, and other natural resources required to produce goods and services, as well as the waste generated by these activities.

4) Life Cycle Assessment (LCA): LCA is a method for evaluating the environmental impact of a product or service throughout its entire life cycle, from raw material extraction to disposal. It can be used to assess the environmental impact of an industry or a country's economic activities.

5) Adjusted Net Savings (ANS): This method evaluates the impact of policies and projects on social and economic welfare, including efficiency, equity, and sustainability. It involves adjusting traditional savings measures to take into account natural resource consumption and environmental damage. Costs associated with these factors are subtracted from the savings rate to obtain a more accurate measure of economic welfare.

5.2 Model establishment

According to the adjusted GDP method, the GGDP formula should be as follow:

$$GGDP = GDP - R - L - M \tag{1}$$

R represents resource consumption value, L represents environmental pollution cost, and M represents environmental governance cost. Next, it is important to find specific and explicit indicators to precisely estimate the value of resource consumption cost, environmental pollution losses, and environmental governance cost.

5.2.1 Computing the value of resource consumption

Following the Integrated Environmental and Economic Accounting (IEEA) framework, the related projects and their classifications on resource and environmental accounting are merged and simplified. According to the formula of resource consumption cost, it is as follows:

$$R = W + O + G + C + F \tag{2}$$

W represents water resource consumption value, O represents oil consumption value, G represents natural gas consumption value, C represents coal consumption value, and F represents forest land consumption value.

5.2.2 Computing environmental pollution loss

The formula for environmental pollution loss considers both direct and indirect losses. Due to the difficulty in accurately estimating many loss indicators that are too small in magnitude, only natural disaster losses caused by environmental pollution are considered as direct losses, while economic losses caused by human health and disease from environmental pollution are considered as indirect economic losses. As economic losses caused by human health and disease from environmental

	Computational methods	Advantages	Disadvantages
Adjusted GDP	deducting the cost of environmental degradation from traditional GDP.	relatively easy to implement, simple to calculate.	the complexity of environmental impacts cannot be fully capture.
Genuine Progress Indicator （GPI）	subtracting the costs of crime, pollution, and other negative externalities from GDP while adding the benefits of positive social factors such as education, healthcare, and others	It is a comprehensive indicator that measures the economic, social, and environmental welfare, captures the value of non-market goods and services, and provides a more comprehensive measure of economic progress.	It is relatively difficult to calculate, subjective, and requires some dependence on expert judgment to assess the value of non-market goods and services.
Ecological Footprint （EF）	measuring the land area and water resources needed to support economic activities	providing more detailed environmental impact information that can be used to identify regions where economic activities are unsustainable	it involves greater resource consumption and may not be applicable to all economic activities
Life Cycle Assessment (LCA)	etermining research objectives, setting boundaries, collecting data, assessing environmental impacts, interpreting results, and communicating findings".	by considering environmental impacts throughout the entire life cycle, it is possible to assess environmental impacts more comprehensively	it is time and resource consuming, and the results often do not account for some non-material environmental impacts
Adjusted Net Savings (ANS)	identifying potential economic and social impact factors; using cost-benefit analysis methods to evaluate the economic benefits of policies or projects; and considering other factors such as fairness and sustainability	providing a more comprehensive measure of economic welfare	it cannot fully capture the value of natural resources
Gross National Net Product (GNNP)	subtracting the cost of natural resource depletion and pollution from GDP, and adding the value of natural capital	this indicator takes into account the damage caused by natural resource depletion and pollution, providing a more accurate picture of economic growth	it fails to capture other important aspects of sustainable development, such as social welfare and income inequality

pollution account for approximately 25% of all losses, this indicator is obtained by multiplying all losses by 25%. Therefore, the formula for environmental pollution loss is as follows:

$$L = D + I \tag{3}$$

In the above formula, D represents the cumulative value of direct economic losses caused by various natural disasters caused by environmental pollution, and I represents the economic losses caused by environmental pollution on human health and disease.

5.2.3 A simplified model of GGDP

Combined with the above analysis, the final construction model of GGDP is as follows:

$$GGDP = GDP - (W + O + G + C + F) - (D + I) - M \tag{4}$$

The symbols represent the following meanings——

R: Resource consumption value

L: Environmental pollution cost

M: Environmental governance cost

W: Water resource consumption value

O: Oil consumption value

G: Natural gas consumption value

C: Coal consumption value

F: Forest land consumption value

D: Economic loss value caused by natural disasters directly related to environmental pollution

I: Economic loss value of human health and disease caused by environmental pollution.

5.3 Comparative analysis of GGDP and GDP

Based on the model constructed in 5.2, we collected the overall data of each indicator in the model globally from 2010 to 2020 and calculated the global GGDP value from 2010 to 2020, which, along with the GDP value during the same period, generated a line chart with time as the horizontal axis and value as the vertical axis.

From the chart, it can be seen that both GDP and GGDP are generally trending upward, and the GGDP curve is approaching the GDP curve, indicating that the portion of green and clean economic growth represented in GDP is increasing. From a global perspective, resource utilization is becoming more efficient, waste is being reduced, and the degree and impact of environmental pollution are also decreasing. This is a very positive trend.

More importantly, the convergence of GGDP with GDP demonstrates the unlimited potential of the economic development model represented by the GGDP indicator. The use of the GGDP

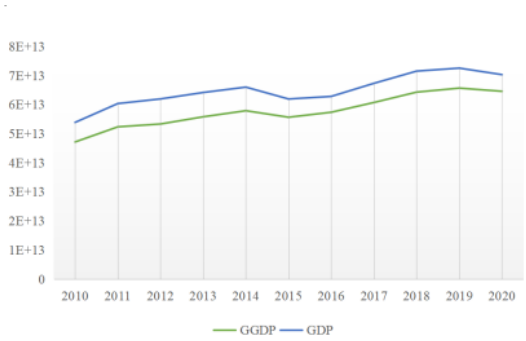


Figure2 GDP&GGDP(current US$)

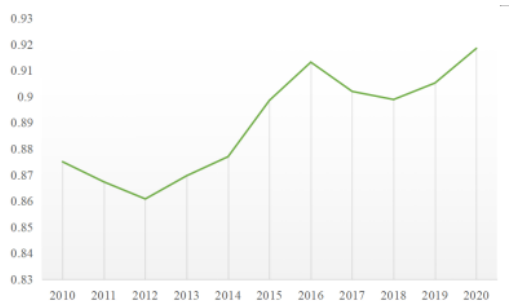


Figure3 The proportion of GGDP in GDP

economic measurement indicator does not necessarily mean that economic development will be restrained.

6 A GGDP-Impact-on-Climate Model

The GGDP index considers not only the contribution of economic growth, but also environmental and sustainable development factors, including factors such as environmental impact, resource utilization efficiency, ecological protection, environmental management, and pollution control, to more comprehensively reflect a country's economic health. The use of GGDP as an indicator to measure economic conditions undoubtedly has an important impact on alleviating current climate change. To specifically measure and compare this impact, we constructed climate indicators and conducted a univariate regression analysis with GGDP and GDP, as well as a multiple regression analysis with variables deducted from GDP during the calculation of GGDP. We obtained important conclusions that the use of GGDP as an indicator to measure economic conditions has an important impact on alleviating current climate change.

6.1 A synthetic indicator of climate mitigation

In order to estimate the expected global impact on climate mitigation, three indicators are selected to represent the climate mitigation. They are:

- Carbon emissions(kt)
- greenhouse gas emissions(kt)
- proportion of renewable energy consumption to total energy consumption

Then we sythesize a composed indicator "climate index"(c_{index}) out of above three indicators, using the method of entropy weight method. Specifically, the establishment c_{index} is as follow:

$$c_{index} = w_1 \cdot M_1 + w_2 \cdot M_2 + w_3 \cdot M_3 \tag{5}$$

The variables in the formula are as follows: M_1 represents carbon emissions, M_2 represents greenhouse gas emissions, M_3 represents the proportion of renewable energy consumption in total

carbon emissions, w_1, w_2, and w_3 represent the weights of the three indicators calculated by the entropy method. In the process of calculating the weights of the entropy method and synthesizing the c_{index}, to eliminate the situation where the optimization directions of the three indicators are not consistent (M_1 and M_2 are better with smaller emissions, while M_3 is better with a higher proportion of renewable energy consumption), the data of M_1 and M_2 were normalized and standardized to eliminate the influence of dimensions. Therefore, the optimization direction of the three indicators is consistent (bigger is better), and the final synthesized c_{index} will also be an indicator where bigger is better.

6.2 Model establishment

To estimate the impact of GGDP as a new economic indicator on climate mitigation, we establish a univariate regression model with GGDP as the independent variable and "climate index" (c_{index}) as the dependent variable as follows:

$$Y_{climatefearure} = \beta_0 + \beta_1 \cdot x_{GGDP} \tag{6}$$

In order to estimate the impact of using GGDP as a new economic measure on climate mitigation, the following simple regression model was established with GGDP as the independent variable and "climate index" (c_{index}) as the dependent variable: Y_cm represents climate mitigation events, expressed by c_{index}, and x_GGDP represents GGDP data. This model is the main model for predicting the impact of GGDP. To compare with the case where GDP is used as the economic measure, we also established the following simple regression model with GDP as the independent variable and c_{index} as the dependent variable:

$$Y_{climatefearure} = \beta_0^* + \beta_1^* \cdot x_{GDP} \tag{7}$$

In the above formula, Y_cm also represents climate mitigation events, represented by c_{index}, and x_GDP represents GDP data. This model will be used to predict the impact of using GDP as an economic measurement indicator on climate mitigation.

6.3 Model Solution

After successfully building the model, we collected annual GDP data for the world from 2010 to 2020, as well as annual data on the value of water consumption, oil consumption, natural gas consumption, coal consumption, forest land consumption, direct economic losses from natural disasters caused by environmental pollution, and economic losses from human health and diseases caused by environmental pollution for the world as a whole from 2010 to 2020. These data have the same unit of measurement, which is US dollars. Afterwards, we calculated the annual values of GGDP from 2010 to 2020 based on the established simplified model of GGDP using the collected data. After obtaining GDP and GGDP data, we collected data on global annual carbon emissions, greenhouse gas emissions, and the proportion of renewable energy consumption in total energy consumption from 2010 to 2020. We then normalized and standardized the data and calculated the weights using the entropy weight method. The weights for carbon emissions, greenhouse gas emissions, and the proportion of renewable energy consumption were 0.34, 0.51, and 0.15, respectively.

$$c_{index} = 0.34M_1 + 0.51M_2 + 0.15M_3 \tag{8}$$

To eliminate the influence of scale, we converted the overall GGDP data to trillions of US dollars and used the Equation(2) in section 6.2 to establish a model for simple linear regression analysis. The final model is as follows:

$$Y_{\text{climate feature}} = 54.88 + (-1.63) \cdot x_{\text{GGDP}} \tag{9}$$

$\beta_0 = 54.88$ and $\beta_1 = -1.63$. According to the model, for every one unit increase in GGDP (in trillions of US dollars), the c_{index}, which measures climate mitigation, will decrease by 1.63 units. This indicates the impact of GGDP on climate mitigation. It can be seen that carbon emissions and greenhouse gas emissions are inevitable when developing the economy, which is understandable.

6.4 Comparison with GDP

After establishing the one-variable statistical regression model between GGDP and the c_{index} that measures climate mitigation, we need to establish a similar model between GDP and c_{index}. To eliminate the influence of GDP's large scale, we also convert GDP's data unit to trillions of US dollars, and perform one-variable statistical regression analysis on it using formula (7) in Section 6.2. The final model is as follows:

$$Y_{\text{climate feature}} = 56.48 + (-1.70) \cdot x_{\text{GDP}} \tag{10}$$

Where $\beta_0 = 56.48$ and $\beta_1 = -1.70$.

According to the model, for every increase of one unit (trillion US dollars) in GDP, the c_{index} will decrease by 1.70 units. Compared with GDP, for every increase of one trillion US dollars in GGDP, it will consume 0.07 fewer units of c_{index}. This fully demonstrates that when GGDP is used as an economic measure, economic growth will significantly reduce carbon and greenhouse gas emissions or significantly increase the proportion of renewable energy use, which undoubtedly benefits climate mitigation. Therefore, using GGDP as an economic measure has a significant positive impact on climate mitigation that cannot be ignored.

The above-calculated two models need to be separately tested for goodness of fit and variable significance. The test results are shown in the table below:

Table 1: Model test table for statistical regression

Statistical regression model	**R-squared**	**t-statistic**
GGDP and c_index	0.9614	-14.1
GDP and c_index	0.9567	-14.96

The table shows that the R-squared values for both models are close to 1, indicating a good fit of the regression models to the sample data. The t-test values indicate that both GDP and GGDP are highly significant variables in their respective models, confirming the linear relationship between

each of them and c-index, and demonstrating the applicability of the simple linear regression model in this context.

7 Potential Upside and Downside During Conversion

7.1 Potential income analysis

According to our previous global impact analysis, it is clear that compared to GDP, GGDP will have less harmful impact on climate mitigation when increased. It could be producing less carbon or greenhouse gas, or having a higher proportion of renewable resource usage. Those differences will definately benefit both the economic and environment in the long run. Using GGDP as an indicator to measure the economic development of a country can motivate governments to actively carry out environmental protection actions and combat behaviors that damage the environment. This will greatly alleviate a series of problems caused by environmental destruction and reduce non-essential but enormous expenditures. For example, due to environmental damage, countries need to invest a large amount of funds each year to solve direct or indirect problems, such as economic losses caused by disasters and losses caused by diseases resulting from environmental pollution.

Excessive industrialization and urbanization not only consume a large amount of energy and natural resources, but also aggravate environmental pollution and ecosystem degradation. This often directly leads to the occurrence of natural disasters, such as floods, droughts, and earthquakes, which pose great risks to people's lives and property. In order to cope with these natural disasters, governments of various countries have to invest a large amount of funds in repair and rescue work. These funds could have been used in more valuable fields, but are now forced to deal with the impact of natural disasters. According to statistics, the total economic losses caused by disasters in various countries around the world have remained above one billion US dollars since 2010, and even reached 49 billion US dollars and 23 billion US dollars in 2016 and 2020, respectively.



Figure5 Economic losses caused by disasters



Figure6 Economic losses caused by diseases

ironmental damage also indicates the urgency of environmental protection. Each year, diseases caused by environmental pollution result in expenditures of up to several hundred billion US dollars worldwide, and this figure has almost consistently risen since 2010. By around 2020, the total expenditure on this item by various countries had reached 25,000 million US dollars.

7.2 Potential cost analysis

However, since 1944, when GDP was formally established as the standard measure of national economy in the Bretton Woods Agreement, GDP has been the recognized measure of global economic development for almost eighty years. Undoubtedly, converting this globally recognized indicator value into GGDP will inevitably result in capital consumption. We classify these potential costs into the following categories:

Cost of data collection and analysis: This includes the cost of collecting, organizing, and analyzing data based on GGDP. For example, government and private sector spending on data collection and processing; indicators of data quality, reliability, and timeliness, such as error rate, omission rate, and data update frequency.

Cost of policy implementation: This includes the cost of adopting policy measures to promote the realization of GGDP. Indicators that may be used to measure this cost include the cost of subsidies, tax reductions, and other policy tools, expenditures related to institutions and human resources involved in policy implementation, and metrics for the efficiency and effectiveness of policy implementation and execution, such as policy feasibility, sustainability, and impact.

Cost of industrial structure adjustment: This includes the cost of shifting the economy from being centered on GDP to being centered on GGDP. This category of costs may include changes in production, investment, and consumption spending across industries during the transformation process, the impact of industrial policies on industrial structural changes, economic growth, innovation, and competitiveness, among others.

Cost of improving resource efficiency: This includes the cost of improving resource efficiency and reducing environmental pollution. Indicators that may be used to measure this cost include the funds and human resources spent on technology innovation and economic policies, efficiency indicators in resource efficiency, energy efficiency, and waste disposal, environmental quality indicators such as air, water, and soil quality.

Among these costs, the potential costs of industrial structural adjustment and resource efficiency improvement in technology and human resource investment are the main two. Meanwhile, the cost of government policy implementation and annual data collection is expected to rise in the short term.

7.3 Comparison and Summary

Although converting GDP to GGDP requires a certain amount of financial investment, adjusting industrial structure and improving production efficiency are inevitable paths for a country to achieve faster and more efficient development. These reforms require a lot of financial investment and relevant infrastructure development, but completing these investments and constructions earlier will provide more development dividends for the country in the future. In addition, the potential benefits of protecting the environment can also be used to further promote the country's economic development and have a positive impact in other areas. For example, implementing environmental protection measures such as clean energy projects and sustainable development projects can create more job opportunities and economic growth opportunities for the country, further promoting industrial upgrading and economic transformation. Therefore, although converting from GDP

to GGDP requires some financial investment, it is a necessary measure that can bring more opportunities and dividends for the country's future economic development, while also promoting economic development while protecting the environment, forming a virtuous cycle.

8 Concrete analysis of switch from GDP to GGDP

8.1 Specific situation in China

China is one of the world's most populous countries and also one of the largest energy and resource-consuming countries in the world. The tight supply of energy and resources has to some extent threatened China's sustainable economic development. In recent years, China's environmental problems have received widespread attention, and pollution in various aspects such as the atmosphere, water bodies, and land has posed a serious threat to people's lives and health. At the same time, these environmental problems have also constrained China's sustainable economic development. From a global perspective, with the increasingly prominent global environmental issues, the international community is paying more and more attention to environmental issues, and environmental regulations and standards are becoming increasingly stringent. As one of the world's largest carbon dioxide emitting countries, China also faces enormous pressure in responding to climate change and international environmental constraints.

In this context, by adopting GGDP, the Chinese government can more comprehensively measure the quality of economic growth and promote China's transition to a more sustainable and green economic model.

8.2 Specific Changes Caused by Switch

To improve GGDP, China should aim to reduce its consumption of natural resources and pollution to the environment while continuously improving labor productivity. Therefore, compared to the past, China may further strengthen the protection of ecosystems, promote sustainable and green urban development models such as green buildings and low-carbon transportation, continuously improve environmental awareness of enterprises and citizens, strengthen environmental monitoring and law enforcement, and reduce environmental damage. Therefore, to achieve these goals, the most important change for China should be to strengthen the implementation of environmental protection policies.

Environmental protection policies can be mainly divided into two types: regulatory policies and subsidy policies, of which regulatory policies can be further divided into two types: investment-based regulations and cost-based regulations. Investment-based regulations typically refer to the government encouraging enterprises to invest in environmental protection facilities through rewards or tax exemptions, to reduce environmental pollution and improve resource utilization efficiency. Cost-based regulations refer to the government imposing environmental pollution fees, waste disposal fees, and other fees on enterprises to reduce environmental pollution and promote resource utilization, establish a sound pollution charging system and regulatory mechanism, and prevent illegal tax evasion by enterprises. Green innovation research and development subsidies refer to the government providing funding and other forms of support to enterprises and research institutions to encourage and promote the research and development of innovative technologies to address environmental problems. This support can take various forms, such as bonuses, subsidies,

tax reductions, scientific research funds, etc. The purpose is to encourage enterprises and research institutions to increase their investment in the environmental field, innovate green technology, improve environmental management and resource utilization efficiency, and promote sustainable development.

In addition, the Chinese government may improve public awareness and consciousness of environmental protection through information disclosure, social participation, and environmental education, and encourage citizens and enterprises to actively participate in environmental and climate protection and sustainable development. Through international cooperation and cross-border governance, China can promote global environmental governance, maintain global ecological security and human common interests.

8.3 Effect analysis

To confirm the impact of China's strengthened environmental policies on promoting the quality and sustainability of economic growth, we selected GTFP as a measurement indicator to analyze the impact of past environmental policies implemented by the Chinese government on it. Therefore, we can predict the expected effects of further strengthening environmental policy implementation. GTFP (Green Total Factor Productivity) is a productivity indicator that evaluates the total value produced by an economic entity or industry over a certain period, taking into account resource utilization efficiency, environmental quality, and production efficiency.

8.3.1 Model Establishment

Constructing a dynamic model to examine the effects of environmental policy tools on green total factor productivity.

$$GTFP_{it} = \alpha_0 + \alpha_1 \cdot lnER_{it} + \alpha_2 \cdot lnSub_{it} + \lambda \cdot X_{it} + CV_{it} + f_i + f_t + \epsilon_{it} \tag{11}$$

In this model, the coefficients α_1 and α_2 represent the total effects of regulatory policies and subsidy policies on green total factor productivity (GTFP), a comprehensive productivity index that evaluates the total output value of a certain economic entity or industry over a certain period of time, taking into account resource utilization efficiency, environmental quality, and production efficiency. $GTFP_{it}$=[$SGTFP_{it}, MGTFP_{it}, LGTFP_{it}$], where $SGTFP_t$, $MGTFP_t$, and $LGTFP_t$ represent the short-term, medium-term, and long-term levels of GTFP for a specific province in year t+1, respectively. If Y_{it} represents the measured value of GTFP for province i in year t, then $SGTFP_t = Y_t + 1$, $MGTFP_t = (Y_t + 1 + Y_t + 2 + Y_t + 3)/3$, and $LGTFP_t = (Y_t + 1 + Y_t + 2 + ... + Y_t + 5)/5$. Due to the lag effect of environmental policy tools on GTFP, SGTFP, MGTFP, and LGTFP are all calculated from the leading one-period item of Y. X_{it} represents the lagged variable set, λ represents the coefficient set of X_{it}. CV represents the control variable and its coefficient set, including labor force scale (Labor), physical capital (Capital), industry structure (Structure), domestic enterprise profitability (Roe), and pollutant discharge intensity (PI). f_i represents individual characteristics that do not vary over time, f_t represents year effects, and $\epsilon_i t$ represents random disturbances.

8.3.2 Estimation Method, Data Sources, and Indicator Processing

In this study, we use the System GMM estimation method to estimate the equation mentioned above. The sample is based on the panel data of provincial-level industry in China from 2006 to 2019, excluding data from Tibet and the Hong Kong, Macau, and Taiwan regions. The sample data are obtained from the "China Statistical Yearbook," "China Environmental Yearbook," "China Industrial Statistical Yearbook," "China Science and Technology Statistical Yearbook," "China Energy Statistical Yearbook," and the National Bureau of Statistics website. Some missing data for individual years were obtained through differentiation. The specific construction methods for the variable indicators are as follows:

- **The bath water is incompressible Non-Newtonian fluid**. Green Total Factor Productivity (GTFP). Following the method of Molinos-Senante et al. (2014), a Global Malmquist-Luenberger productivity index is constructed to measure the green total factor productivity at the regional (provincial) level.
- **Dependent variable:**.Environmental policy instruments. The investment-oriented regulatory intensity (IER) is measured by the ratio of environmental protection investment in "three simultaneities" construction projects to the production value of industrial enterprises above designated size in the region. The cost-based environmental regulatory intensity (FER) is measured by the ratio of pollutant discharge fees collected to the number of paying companies. The research and development subsidy intensity (Sub) is measured by the internal expenditure of government-funded RD for industrial enterprises.
- **Core explanatory variables:**. Before the person lies in the bathtub, no internal heat source exist except the system components. The circumstance where the person is in the bathtub will be investigated in our later discussion.
- **Control variables:**. Industrial structure (Structure) is measured by the ratio of the production value of industrial enterprises above designated size to the regional gross domestic product. Labor scale (Labor) is measured by the average number of employees per year. Physical capital (Capita) is measured by the perpetual inventory method of capital stock. Profitability of local enterprises (Roe) is measured by the return on equity. Pollution intensity (PI) is calculated by constructing a composite index based on the amount of industrial wastewater, waste gas, and solid waste emissions in the region.

8.3.3 Analysis of empirical results

The results of the test for model adequacy indicate that when short-term green total factor productivity (GTFP) is used as the dependent variable, the first and second lagged dependent variables should both be included in the model. When medium-term and long-term GTFP are used as the dependent variables, only the first lagged dependent variable needs to be included in the model. The Sargan test results are not significant, indicating that the instrumental variables used in the model are effective. Table 4 shows that a 1% increase in investment-oriented regulatory intensity leads to a 0.068 increase in GTFP in the short term, a 0.020 decrease in the medium term, and a 0.035 decrease in the long term. This suggests that the negative effect of investment-oriented

Table 2: Descriptive statistics of variables

VARIABLE	NUM	MEAN	STA	MIN	MAX	MID
GTFP	360	1.104	0.201	0.632	1.991	1.059
LNLER	360	-5.864	0.845	-8.575	-1.986	-5.933
INFER	360	1.605	0.678	0.079	3.428	1.624
LNSUB	360	10.953	1.281	6.273	13.067	11.12
LNLABOR	360	5.174	1.102	2.084	7.308	5.169
INCAPITAL	360	8.879	0.832	6.269	10.654	8.968
LNPI	360	1.115	0.375	0.26	2.84	1.068
STRUCTURE	360	0.392	0.084	0.118	0.564	0.408
ROE	360	0.069	0.028	0.003	0.198	0.065

regulation on GTFP increases over time. A 1% increase in cost-oriented regulatory intensity leads to a 0.211 decrease in GTFP in the short term, a 0.026 decrease in the medium term, and a 0.039 increase in the long term. This indicates that the positive effect of cost-oriented regulation on GTFP increases over time. The intensity of research and development subsidies has a positive effect on GTFP in the short, medium, and long terms, and the positive effect increases over time.

Table 3: Empirical analysis results

	SGTFP	MGTFP	LGTFP
lnIER	0.068***	-0.020***	-0.035***
	(-0.0092)	(-0.0053)	(-0.0035)
lnFER	-0.211***	-0.026**	0.039**
	(-0.0254)	(-0.0131)	(-0.0172)
lnSub	0.028***	0.040***	0.054***
	(-0.0149)	(-0.0051)	(-0.0056)
N	330	300	240
L1GTFP	0.183***	0.530***	0.401***
	(-0.0255)	(-0.0471)	(-0.0385)
L2GTFP	-0.455***	-	-
	(-0.0263)		
Sargan test	27.921	27.964	23.531
Control variable	YES	YES	YES
Method of estimation	SGMM	SGMM	SGMM

8.4 Conclusion

Therefore, when China adopts GGDP as a measure of economic development, it can promote green total factor productivity in both the short and long term through the implementation of a combination of three policy tools. This can reduce the consumption of natural resources and environmental impact while promoting the development of green economy and innovation, thereby promoting China's green transformation. This will help to mitigate environmental pollution,

ecological damage and other environmental problems, improve the efficiency of natural resource utilization, promote the construction of ecological civilization, protect the ecological environment, and improve the health level of the people.

Secondly, the improvement of green total factor productivity can promote China's sustainable economic development. This is because, by improving production efficiency and resource utilization efficiency, it can reduce the dependence of economic development on resources and the environment, promote the quality change of economic growth, and achieve the sustainability of economic development. At the same time, the improvement of green total factor productivity can also promote technological progress and technological innovation, thereby promoting industrial upgrading and transformation, improving China's core competitiveness, and conducive to maintaining long-term stable economic growth.

In conclusion, this is a long-term process that requires the joint efforts of the government, enterprises and the public. If successfully implemented, it will have a positive impact on China's economic and social sustainable development, and also contribute to the harmonious coexistence of humans and nature.

9 Model Test——Robustness test and Stress test

To ensure the reliability of the regression results for Equation (11), robustness tests need to be conducted. First, a variable selection bias test is performed by using alternative indicators to measure explanatory variables. The proportion of enterprises participating in clean production audits in each province as a percentage of the total number of industrial enterprises above designated size is used as a proxy for the intensity of investment-type regulation, and the ratio of environmental administrative fines and confiscations to the operating costs of industrial enterprises above designated size in each province is used as a proxy for the intensity of cost-type regulation. The regression results are shown in columns (1) to (3) of Table 4.

Table 4: Model test result

	Variable selection bias test			**Model pressure test**		
	SGTFP	MGTFP	LGTFP	SGTFP	MGTFP	LGTFP
lnIER	0.012**	-0.002	-0.017***	0.043***	-0.008*	-0.011***
	(-0.0065)	(-0.0026)	(-0.0056)	(-0.0079)	(-0.0043)	(-0.0035)
lnFER	-0.035*	0.001	0.095***	-0.093**	-0.142***	0.129***
	(-0.0194)	(-0.0195)	(-0.045)	-0.0403)	(-0.0155)	(-0.025)
lnSub	0.037**	0.048***	0.056***	-0.012	0.104***	0.096***
	(-0.0167)	(-0.009)	(-0.0052)	(-0.0209)	(-0.0062)	(-0.0056)
N	330	300	240	300	270	210
L1GTFP	0.142***	0.578***	0.412***	0.175***	0.554***	0.461***
	(-0.0415)	(-0.0366)	(-0.035)	(-0.0309)	(-0.0392)	(-0.0336)

Secondly, a model stress test is conducted to test the fit between the empirical model and the sample data. The difference generalized method of moments estimation method (DGMM) is used to perform the model stress test on equation (11): on the one hand, using DGMM means that fewer samples can participate in the regression compared to SGMM; on the other hand, the DGMM method eliminates the f variable in the equation that does not vary with time, relaxing the constraints

of the model on the random disturbance term. The results of the model stress test are shown in columns (4) to (6) of Table 4.

Examining the results of the robustness tests, it is found that the trends in the signs and magnitudes of the coefficients of the main explanatory variables remain consistent with the results in Table 3 as the values of the dependent variable change (the time window extends), and the significance levels remain mostly consistent with the original regression results. This indicates that the regression results of equation (11) are relatively reliable.

Therefore, it can be concluded that investment-type regulation in environmental policies has a significant promoting effect on green total factor productivity in the short term, while cost-type regulation and green innovation research and development subsidies have a strong long-term promoting effect on green total factor productivity.

10 Model Evaluation

10.1 Strength

- We analyze the problem based on thermodynamic formulas and laws, so that the model we established is of great validity.
- Our model is fairly robust due to our careful corrections in consideration of real-life situations and detailed sensitivity analysis.
- Via Fluent software, we simulate the time field of different areas throughout the bathtub. The outcome is vivid for us to understand the changing process.
- We come up with various criteria to compare different situations, like water consumption and the time of adding hot water. Hence an overall comparison can be made according to these criteria.
- Besides common factors, we still consider other factors, such as evaporation and radiation heat transfer. The evaporation turns out to be the main reason of heat loss, which corresponds with other scientist's experimental outcome.

10.2 Weakness

- Having knowing the range of some parameters from others' essays, we choose a value from them to apply in our model. Those values may not be reasonable in reality.
- Although we investigate a lot in the influence of personal motions, they are so complicated that need to be studied further.
- Limited to time, we do not conduct sensitivity analysis for the influence of personal surface area.

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